

Exp No. 2: Implementing TDMA Slot Allocation in GSM.

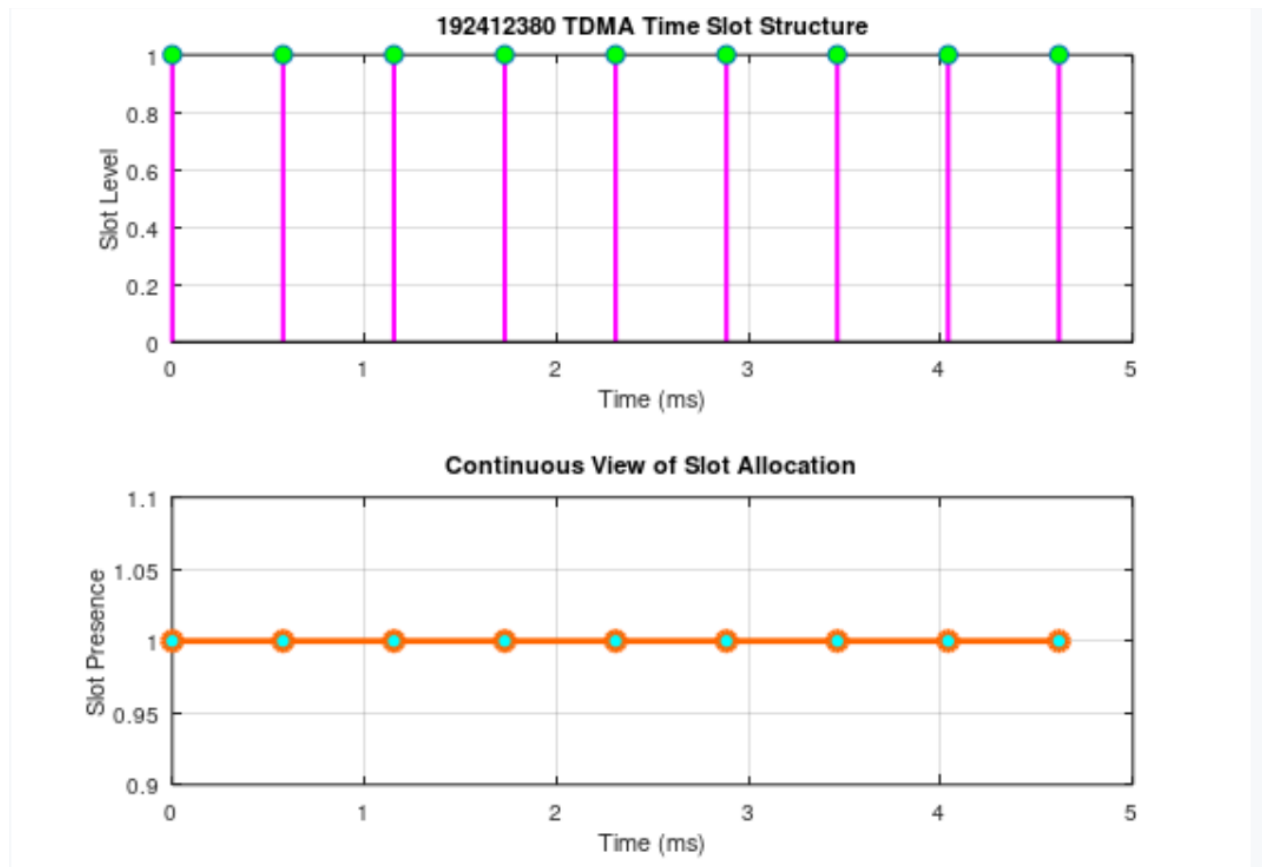
AIM: To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

Objective 1: To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

Program

```
1  clc;
2  clear;
3  close all;
4
5  % TDMA Parameters
6  FrameTime = 4.615;      % Frame Time (ms)
7  Slots = 8;             % Number of Time Slots
8  SlotDuration = FrameTime / Slots;
9
10 % Time Vector
11 t = 0:SlotDuration:FrameTime;
12
13 % First Plot - Time Slot Structure
14 subplot(2,1,1);
15
16 stem(t, ones(size(t)), ...
17      'filled', ...
18      'Color', 'm', ...
19      'MarkerFaceColor', 'g', ...
20      'Linewidth', 1.5);
21
22 title('192412380 TDMA Time Slot Structure');
23 xlabel('Time (ms)');
24 ylabel('Slot Level');
25 grid on;
26
27 % Second Plot - Continuous Slot Allocation
28 subplot(2,1,2);
29
30 plot(t, ones(size(t)), ...
31      '-o', ...
32      'Color', [1 0.4 0], ...
33      'MarkerFaceColor', 'c', ...
34      'Linewidth', 2);
```

OUTPUT:



Objective 2: To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

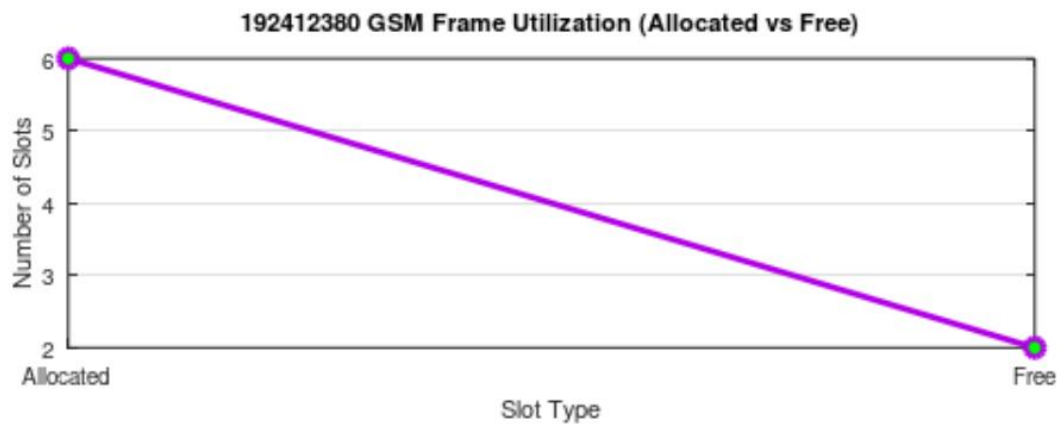
Program

```

1  clc;
2  clear;
3  close all;
4
5  % GSM Frame Parameters
6  TotalSlots = 8;
7  AllocatedSlots = 6;
8  FreeSlots = TotalSlots - AllocatedSlots;
9
10 % Frame Utilization
11 Utilization = (AllocatedSlots / TotalSlots) * 100;
12
13 % Display Result
14 fprintf('Frame Utilization = %.2f %%\n', Utilization);
15
16 figure;
17
18 %----- Graph 1 -----%
19 subplot(2,1,1);
20
21 plot([1 2], [AllocatedSlots FreeSlots], ...
22      '-o', ...
23      'Color', [0.8 0 0.8], ...      % Purple line
24      'MarkerFaceColor', [0 1 0], ...% Green marker
25      'Linewidth', 2);
26
27 title('GSM Frame Utilization (Allocated vs Free)');
28 xlabel('Slot Type');
29 ylabel('Number of Slots');
30
31 set(gca,'XTick',[1 2]);
32 set(gca,'XTickLabel',{'Allocated','Free'});
33
34 grid on;

```

OUTPUT:



Slot Distribution in GSM Frame

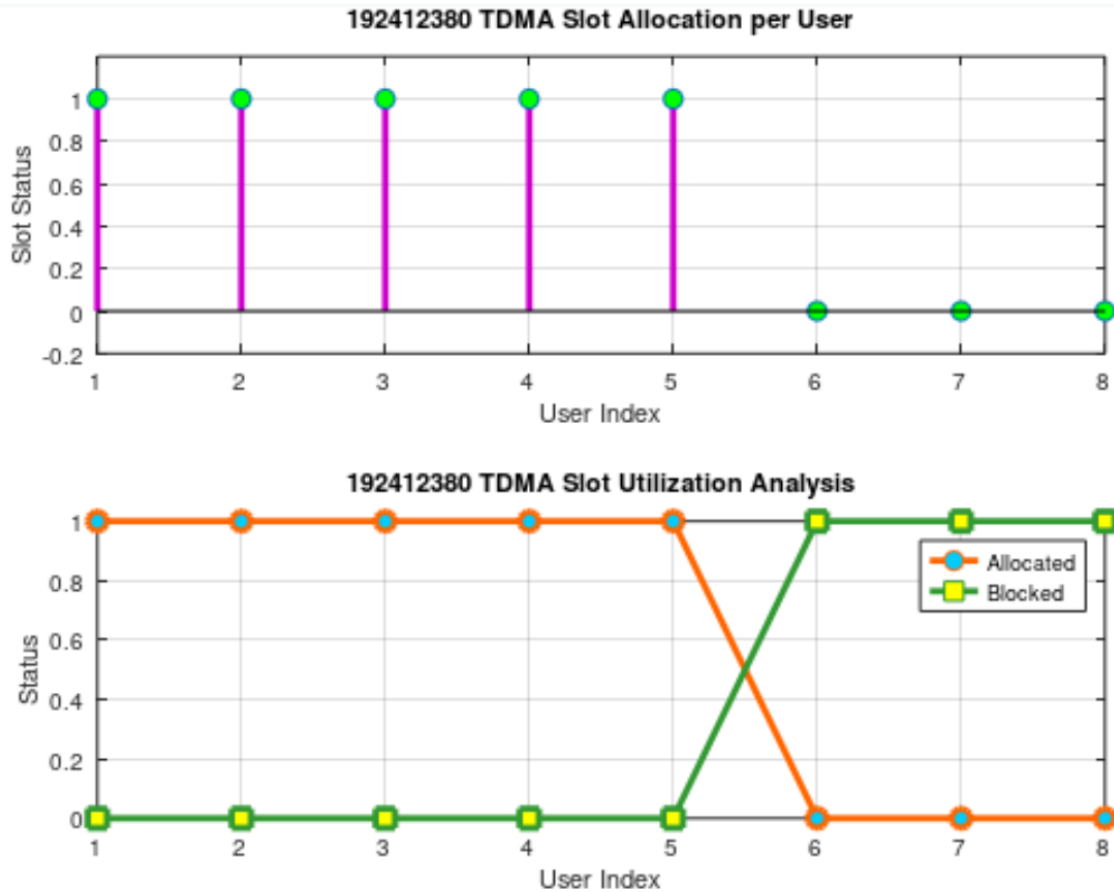


Objective 3: To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

Program:

```
1  clc;
2  clear;
3  close all;
4
5  % TDMA Slot Allocation
6  Users = 1:8;          % Total Users
7  TotalSlots = 5;       % Available Slots
8
9  % Slot Allocation
10 Alloc = zeros(1,8);
11 Alloc(1:TotalSlots) = 1;
12
13 % Blocked Users
14 Blocked = 1 - Alloc;
15
16 % Display Results
17 fprintf('User\tStatus\n');
18 fprintf('-----\n');
19 for i = 1:length(Users)
20     fprintf('%d\t%d\n', Users(i), Alloc(i));
21 end
22
23 figure;
24
25 %----- Graph 1 -----%
26 subplot(2,1,1);
27
28 stem(Users, Alloc, ...
29     'filled', ...
30     'Color', 'm', ...
31     'MarkerFaceColor', 'g', ...
32     'Linewidth', 2);
33
34 title('TDMA Slot Allocation per User');
```

OUTPUT:



Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.
2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.
3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.